

Question

In any chemical reaction, equilibrium is supposed to be established when:

- A** Mutually opposite reactions occur.
- B** Concentration of reactants and resulting products are equal.
- C** Velocity of mutual opposite reactions become equal.
- D** The temperature of mutual opposite reactions becomes equal.

Question

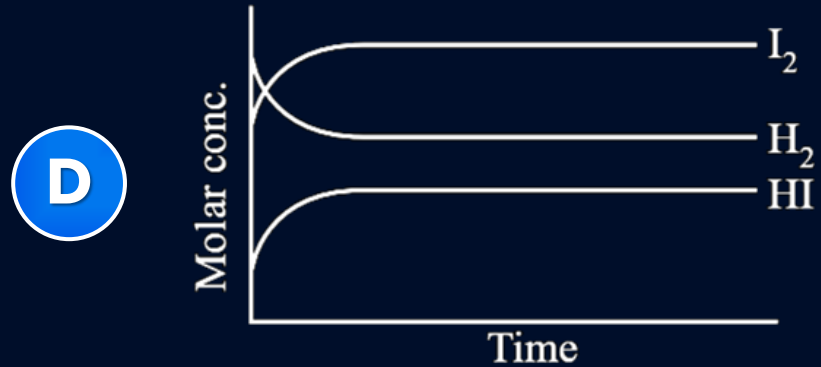
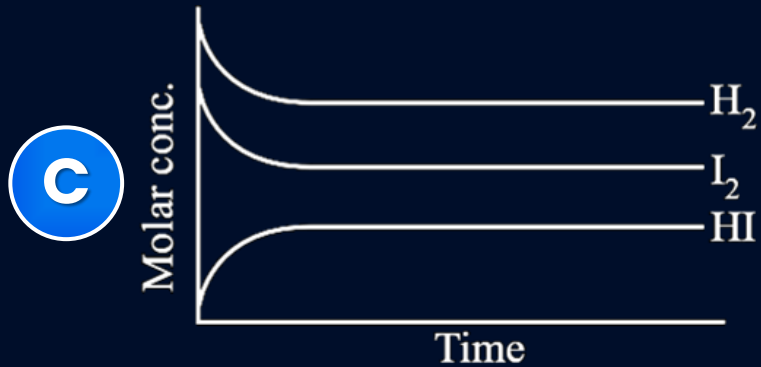
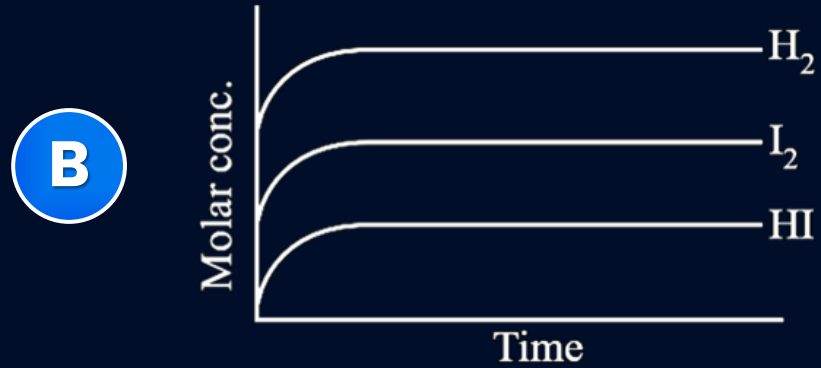
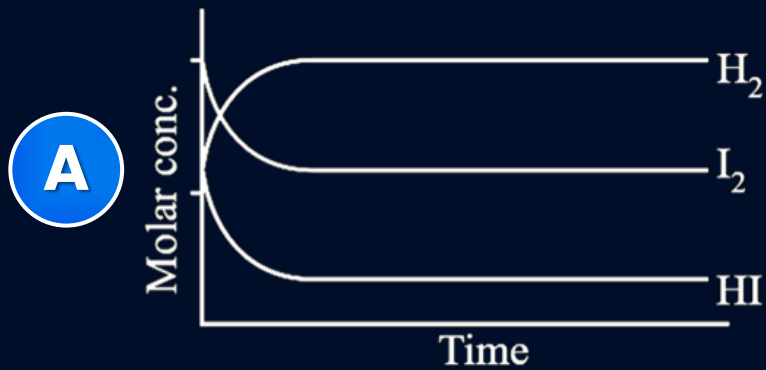
X $\rightleftharpoons$ Y reaction is said to be in equilibrium, when:

- A** Only 10% conversion x to y takes place
- B** Complete conversion of x to y takes place
- C** Only 50% conversion of x to y takes place
- D** The rate of change of x to y is becomes equal to the rate of change of y to x in the system

Question

For the reaction, $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$

Attainment of equilibrium is predicted correctly by:



Question

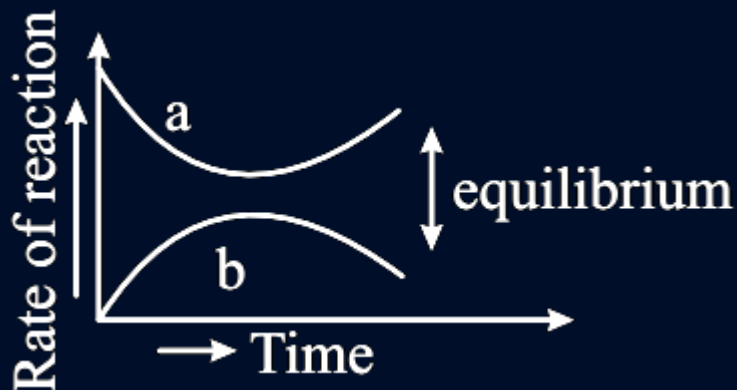
The number of correct statements involving equilibria in physical process from the following is

- A** Equilibrium is possible only in a closed system at a given temperature
- B** Both the opposing processes occur at the same rate
- C** When equilibrium is attained at a given temperature, the value of all its parameters became equal
- D** For dissolution of solids in liquids, the solubility is constant at a given temperature

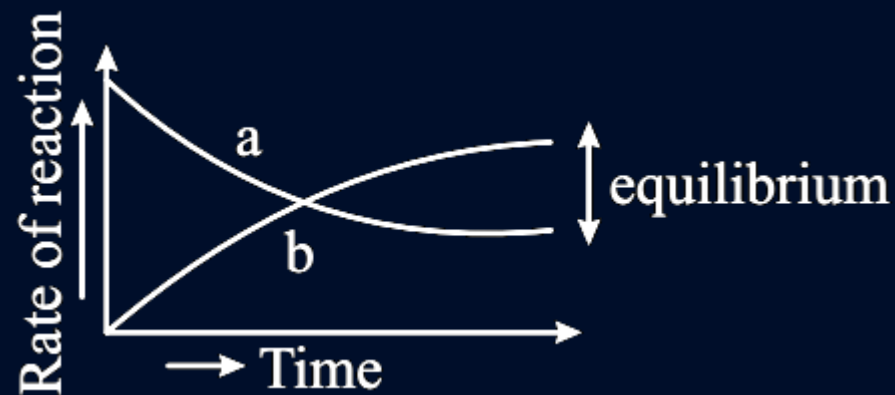
Question

For the equilibrium $A \rightleftharpoons B$, the variation of the rate of the forward (a) and reverse (b) reaction with time is given by:

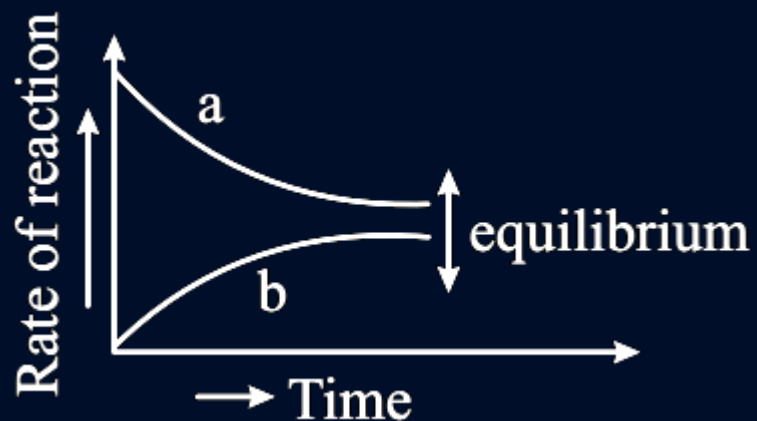
A



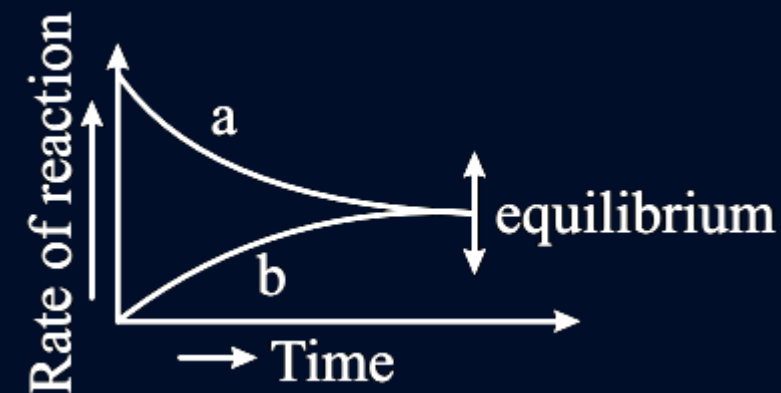
B



C



D



Question

If 8.5 g ammonia is present in a vessel of 0.5 L capacity then find out the active mass of ammonia.

Question

Assertion: The active mass of pure solids and pure liquids is taken as unity.

Reason: The active mass of pure solids and pure liquids depends on density and molecular mass. The density and molecular mass of pure solids and pure liquids are constant.

- A** Both Assertion and Reason are true and Reason is correct explanation of Assertion
- B** Both Assertion and Reason are true and Reason is not correct explanation of Assertion
- C** Assertion is true Reason is false
- D** Reason is true Assertion is false

Question

What should be the active mass in g mol L^{-1} when 0.585 g NaCl is present in 100 mL aqueous solution

- A** 0.1
- B** 0.5
- C** 1.0
- D** 2.0

Question

Calculate the active mass of 40 g NaCl(s)? (NaCl = 58.5)

A 0.68

B 1

C 2

D 4

Question

For a reversible reaction, the rate constant for forward and backward reactions are 2.38×10^{-4} and 8.15×10^{-5} respectively. The equilibrium constant for the reaction is:

- A** 0.342
- B** 2.92
- C** 0.292
- D** 3.42

Question

In a chemical equilibrium, the rate constant for the backward reaction is 7.5×10^{-4} and the equilibrium constant is 1.5. The rate constant for the forward reaction is:

- A** 2×10^{-3}
- B** 5×10^{-4}
- C** 1.12×10^{-3}
- D** 9.0×10^{-4}

Question

At 3000 K, the equilibrium pressures of CO_2 , CO and O_2 are 0.6, 0.4 and 0.2 atm respectively. K_p for the reaction, $2\text{CO}_2 \rightleftharpoons 2\text{CO} + \text{O}_2$ is:

- A** 0.089
- B** 0.0533
- C** 0.133
- D** 0.177

Question

On a given condition, the equilibrium concentration of HI, H₂ and I₂ are 0.80, 0.10 and 0.10 mole/litre. The equilibrium constant for the reaction will be



A 64

B 12

C 8

D 0.8

Question

A quantity of PCl_5 was heated in a 10 litre vessel at 250°C ; $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$. At equilibrium the vessel contains 0.1 mole of PCl_5 , 0.2 moles of PCl_3 and 0.2 moles of Cl_2 . The equilibrium constant of the reaction is:

A 0.02

B 0.05

C 0.04

D 0.025

Question

For the reaction $2\text{C}_{(s)} + \text{CO}_{2(g)} \rightarrow 2\text{CO}_{(g)}$ the partial pressure of CO_2 and CO are 2.0 and 4.0 atm respectively at equilibrium. The K_p for the reaction is

A 0.5

B 4.0

C 8.0

D 32.0

Question

In the reaction, $A + B \rightleftharpoons 2C$, at equilibrium, the concentration of A and B is 0.20 mol L^{-1} each and that of C was found to be 0.60 mol L^{-1} . The equilibrium constant of the reaction is

A 2.4

B 18

C 4.8

D 9

Question

For the equilibrium $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$, K_c at 1000K is 2.37×10^{-3} . If at equilibrium $[\text{N}_2] = 2\text{M}$, $[\text{H}_2] = 3\text{M}$, the concentration of NH_3 is:

- A** 0.00358 M
- B** 0.0358 M
- C** 0.358 M
- D** 3.58 M

Question

8 g hydrogen is present in a vessel of 1 L capacity, Find out the active mass of hydrogen.

- A** 1 mol L^{-1}
- B** 4 mol L^{-1}
- C** 0.5 mol L^{-1}
- D** 2.5 mol L^{-1}

Question

What should be the active mass in mol L^{-1} when 0.745 g KCl is present in 1000 mL solution?

- A** 0.5
- B** 0.1
- C** 2
- D** 0.01

Question

The equilibrium constant expression for the equilibrium $2\text{NH}_3(\text{g}) + 2\text{O}_2(\text{g}) \rightleftharpoons \text{N}_2\text{O}(\text{g}) + 3\text{H}_2\text{O}(\text{g})$ is

A $K_{\text{C}} = \frac{[\text{N}_2\text{O}][\text{H}_2\text{O}]^3}{[\text{NH}_3][\text{O}_2]}$

B $K_{\text{C}} = \frac{[\text{H}_2\text{O}]^3[\text{N}_2\text{O}]}{[\text{NH}_3]^2[\text{O}_2]^2}$

C $K_{\text{C}} = \frac{[\text{NH}_3]^2[\text{O}_2]^2}{[\text{N}_2\text{O}][\text{H}_2\text{O}]^3}$

D $K_{\text{C}} = \frac{[\text{NH}_3][\text{O}_2]}{[\text{N}_2\text{O}][\text{H}_2\text{O}]}$

Question

An equilibrium mixture of the reaction $2\text{H}_2\text{S}(\text{g}) \rightleftharpoons 2\text{H}_2(\text{g}) + \text{S}_2(\text{g})$ has 0.5 mole H_2S , 0.10 mole H_2 and 0.4 mole S_2 in one litre vessel. The value of equilibrium constant (K) in mole litre^{-1} is

A 0.004

B 0.008

C 0.016

D 0.160

Question

The following concentrations were observed at 500 K for the formation of NH_3 from N_2 and H_2 . At equilibrium: $[\text{N}_2] = 2 \times 10^{-2} \text{ M}$, $[\text{H}_2] = 3 \times 10^{-2} \text{ M}$ and $[\text{NH}_3] = 1.5 \times 10^{-2} \text{ M}$. Equilibrium constant for the reaction is _____.

Question

For the given reaction, choose the correct expression of K_c from the following:



A $K_c = \frac{[\text{FeSCN}^{2+}]}{[\text{Fe}^{3+}][\text{SCN}^{-}]}$

B $K_c = \frac{[\text{Fe}^{3+}][\text{SCN}^{-}]}{[\text{FeSCN}^{2+}]}$

C $K_c = \frac{[\text{FeSCN}^{2+}]}{[\text{Fe}^{3+}]^2[\text{SCN}^{-}]^2}$

D $K_c = \frac{[\text{FeSCN}^{2+}]^2}{[\text{Fe}^{3+}][\text{SCN}^{-}]}$

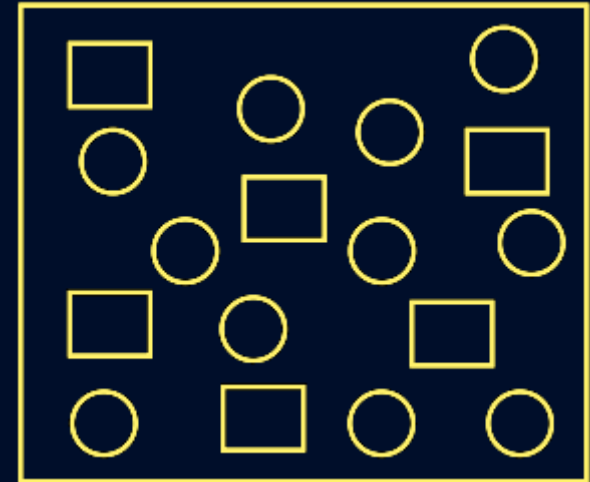
Question

The equilibrium constant for the reaction $A(s) \rightleftharpoons M(s) + \frac{1}{2} O_2(g)$ is $K_p = 4$. At equilibrium, the partial pressure of O_2 is _____ atm. (Round off to the Nearest Integer).

Question

If the figure shown below reactant A (represented by square) is in equilibrium with product B (represented by circle). The equilibrium constant is:

- A** 8
- B** 2
- C** 4
- D** 1



Question



In an equilibrium mixture, the partial pressures are

$P_{\text{SO}_3} = 43\text{kPa}$; $P_{\text{O}_2} = 530\text{ Pa}$ and $P_{\text{SO}_2} = 45\text{kPa}$. The equilibrium constant $K_p =$ _____ $\times 10^{-2}$ (Nearest integer)

Question



Based on above equilibria, the equilibrium constant of the reaction At 298 K,

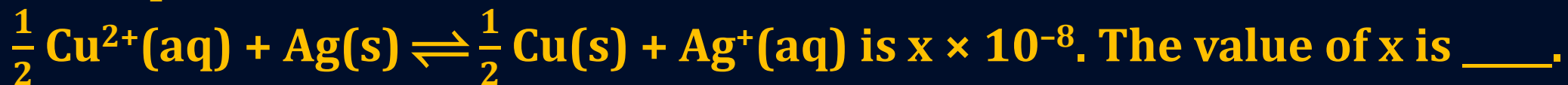


Question

At 298 K, the equilibrium constant is 2×10^{15} for the reaction:



The equilibrium constant for the reaction



(Nearest Integer)

Question

The value of K_c is 64 at 800 K for the reaction $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$.

The value of K_c for the following reaction is:



A 1/64

B 8

C 1/4

D 1/8

Question

A reversible reaction having two reactants in equilibrium if the concentration of reactants are doubled, the equilibrium constant will:

- A** Become 4 times
- B** Become $\frac{1}{4}$ th times
- C** Become $\frac{1}{4}$ th times
- D** Remains the same

Question

The equilibrium constant for the reaction: $\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{NO}(\text{g})$ at 2000 K 4×10^4 . In presence of catalyst the equilibrium is established ten times faster the same temperature. What is the value of equilibrium constant in presence catalyst:

- A** 40×10^{-4}
- B** 4×10^{-4}
- C** 4×10^4
- D** None

Question

For the reaction $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$, $K_p = 0.492$ atm at 300K.

K_c for the reaction at same temperature is $___ \times 10^{-2}$.

(Given: $R = 0.082 \text{ L atm mol}^{-1} \text{ K}^{-1}$)

Question

For reaction: $\text{SO}_2(\text{g}) + \frac{1}{2} \text{O}_2(\text{g}) \rightleftharpoons \text{SO}_3(\text{g})$

$K_p = 2 \times 10^{12}$ at 27°C and 1 atm pressure. The K_c for the same reaction is _____ $\times 10^{13}$. (Nearest integer) (Given $R = 0.082 \text{ L atm K}^{-1} \text{ mol}^{-1}$)

Question

Value of K_p for the equilibrium reaction $\text{N}_2\text{O}_{4(g)} \rightleftharpoons 2\text{NO}_{2(g)}$ at 288 K is 47.9. The K_c for this reaction at same temperature is _____.

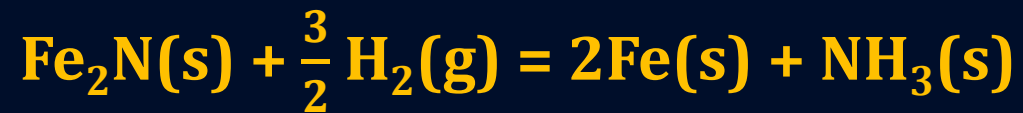
(Nearest integer) $[R = 0.083 \text{ L bar K}^{-1} \text{ mol}^{-1}]$

Question

Consider the reaction $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$. The temperature at which $K_c = 20.4$ and $K_p = 600.1$, is _____ K. (Round off to the nearest integer).
[Assume all gases are ideal and $R = 0.0831 \text{ L bar mol}^{-1} \text{ K}^{-1}$]

Question

For the reaction



- A** $K_c = K_p(RT)$
- B** $K_c = K_p(RT)^{-1/2}$
- C** $K_c = K_p(RT)^{1/2}$
- D** $K_c = K_p(RT)^{3/2}$

Question

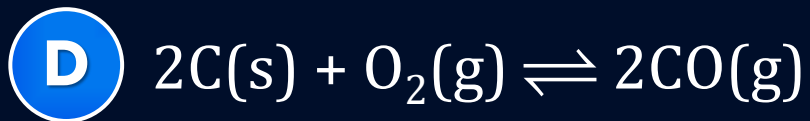
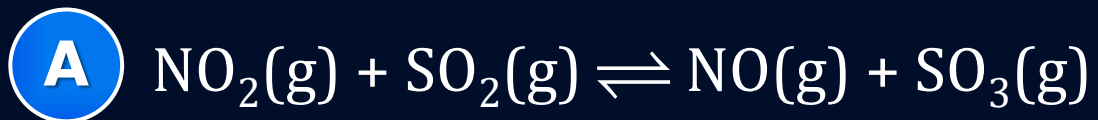
The value of K_p/K_c for the following reactions at 300 K are respectively:
(At 300 K, $RT \rightleftharpoons 24.62 \text{ dm}^3 \text{ atm mol}^{-1}$)



- A** 1, $24.62 \text{ dm}^3 \text{ atm mol}^{-1}$, $606.0 \text{ dm}^6 \text{ atm}^2 \text{ mol}^{-2}$
- B** 1, $24.62 \text{ dm}^3 \text{ atm mol}^{-1}$, $1.65 \text{ dm}^{-6} \text{ atm}^{-2} \text{ mol}^2$
- C** 1, $4.1 \times 10^{-2} \text{ dm}^{-3} \text{ atm}^{-1} \text{ mol}$, $606.0 \text{ dm}^6 \text{ atm}^2 \text{ mol}^{-2}$
- D** $24.62 \text{ dm}^3 \text{ atm mol}^{-1}$, $606.0 \text{ dm}^6 \text{ atm}^2 \text{ atm}^2 \text{ mol}^{-2}$, $1.65 \times 10^{-3} \text{ dm}^{-6} \text{ atm}^{-2} \text{ mol}^2$

Question

In which one of the following equilibrium, $K_p \neq K_c$?



Question

When the two reactions occur at the same rate, the system reaches

- A** a state of completion
- B** a state of equilibrium
- C** 90% completion
- D** 50% completion

Question

The equilibrium constant of the equilibrium equation $2\text{NH}_3 \rightleftharpoons \text{N}_2(\text{g}) + 3\text{H}_2$ is 0. at 1100 K. The value of equilibrium constant $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$ will be:

A 2

B 1

C 4

D 3

Question

If the equilibrium constant of the reversible reaction $\text{HI}_{(\text{g})} \rightleftharpoons \frac{1}{2} \text{H}_{2(\text{g})} + \frac{1}{2} \text{I}_{2(\text{g})}$ is 7.4, the equilibrium constant for the reversible reaction $2\text{HI}_{(\text{g})} \rightleftharpoons \text{H}_{2(\text{g})} + \text{I}_{2(\text{g})}$ will be—

- A** $\sqrt{7.4}$
- B** 54.76
- C** 14.8
- D** 7.4

Question

For the reactions:



K_c for the reaction $A \rightleftharpoons D$:

A 12

B $4/3$

C 24

D 48

Question

The equilibrium constant expression for a gas reaction is,

$$K_c = \frac{[\text{NH}_3]^4 [\text{O}_2]^5}{[\text{NO}]^4 [\text{H}_2\text{O}]^6}$$

Write the balanced chemical equation corresponding to this expression.

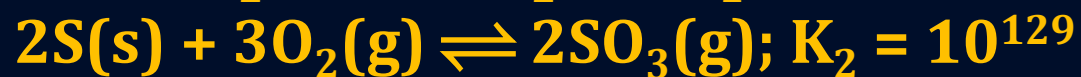
Question

If the equilibrium constant for $A \rightleftharpoons B + C$ is $K^{(1)}_{(eq)}$ and that of $B + C \rightleftharpoons P$ is $K^{(2)}_{(eq)}$, the equilibrium constant for $A \rightleftharpoons P$ is:

- A** $K^{(1)}_{(eq)} K^{(2)}_{(eq)}$
- B** $K^{(1)}_{(eq)} / K^{(2)}_{(eq)}$
- C** $K^{(1)}_{(eq)} + K^{(2)}_{(eq)}$
- D** $K^{(2)}_{(eq)} - K^{(1)}_{(eq)}$

Question

For the following reactions, equilibrium constants are given:



The equilibrium constant for the reaction, $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$ is:

A 10^{181}

B 10^{154}

C 10^{25}

D 10^{77}

Question

Value of K_p for the equilibrium reaction $\text{N}_2\text{O}_{4(g)} \rightleftharpoons 2\text{NO}_{2(g)}$ at 288 K is 47.9. The K_c for this reaction at same temperature is _____.

(Nearest integer) $[R = 0.083 \text{ L bar K}^{-1} \text{ mol}^{-1}]$

Question

For the reaction $\text{CO(g)} + \text{Cl}_2\text{(g)} \rightleftharpoons \text{COCl}_2\text{(g)}$, the $\frac{K_p}{K_c}$ is equal to:

- A** $\frac{1}{RT}$
- B** RT
- C** $\sqrt{RT}$
- D** 1

Question

For the reaction $2\text{NO}_{2(g)} \rightleftharpoons 2\text{NO}_{(g)} + \text{O}_{2(g)}$ ($K_c = 1.8 \times 10^{-6}$ at 184°C),
($R = 0.0831 \text{ kJ}/(\text{mol} \cdot \text{K})$)

When K_p and K_c are compared at 184°C it is found that

- A** K_p is greater than K_c
- B** K_p is less than K_c
- C** $K_p = K_c$
- D** Whether K_p is greater than, less than or equal to K_c depends upon the total gas pressure

Question

At 1000 K, the value of K_p for the reaction:

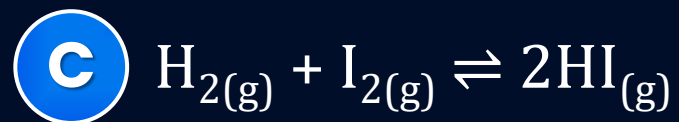
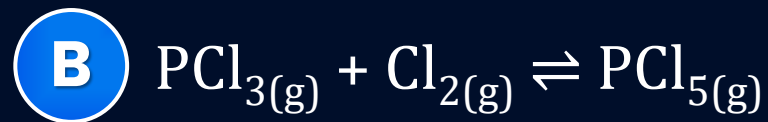
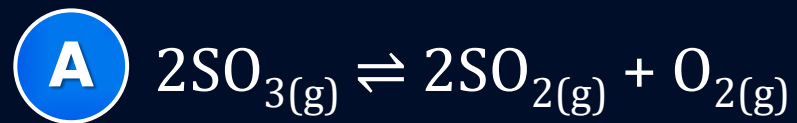


is 0.5 atm. The value of K_c in terms of R would be

- A** 20000 R
- B** 0.02 R
- C** 5×10^{-5} R
- D** $5 \times 10^{-5} R^{-1}$

Question

In which of the following equilibrium equation, $K_p > K_c$?



Question

For homogeneous gas reaction



The equilibrium constant K_c has the unit of

- A** (concentration)¹
- B** (concentration)⁻¹
- C** (concentration)⁹
- D** (concentration)¹⁰

Question

For the reaction $2\text{NH}_3 \rightleftharpoons \text{N}_{2(g)} + 3\text{H}_{2(g)}$ the unit of K_p will be-

A $(\text{atm})^{-2}$

B $(\text{atm})^2$

C $(\text{atm})^3$

D atm

Question

The role of a catalyst in a chemical reaction is:

- A** To help attain equilibrium in a short time.
- B** To affect the state of equilibrium
- C** To shift the equilibrium in such a way so as to increase the concentration of the product
- D** Both (A) and (B)

Question

Assertion: In the presence a catalyst, the value of equilibrium constant increases.

Reason: Catalyst increases the rate of forward and backward reaction the same extent.

- A** Both Assertion and Reason are true and Reason is correct explanation of Assertion
- B** Both Assertion and Reason are true and Reason is not correct explanation of Assertion
- C** Assertion is true Reason is false
- D** Reason is true Assertion is false

Question

If the value of an equilibrium constant for a particular reaction is 1.6×10^{12} , then at equilibrium the system will contain

- A** All reactants
- B** Mostly reactants
- C** Mostly products
- D** Similar amounts of reactants and products

Question

Which oxide of Nitrogen is most stable:

- A** $2\text{NO}_{2(g)} \rightleftharpoons \text{N}_{2(g)} + 2\text{O}_{2(g)}$; $K = 6.7 \times 10^{16} \text{ mol L}^{-1}$
- B** $2\text{NO}_{(g)} \rightleftharpoons \text{N}_{2(g)} + \text{O}_{2(g)}$; $K = 2.2 \times 10^{30}$
- C** $2\text{N}_2\text{O}_{5(g)} \rightleftharpoons 2\text{N}_{2(g)} + 5\text{O}_{2(g)}$; $K = 1.2 \times 10^{34} \text{ mol}^5 \text{ L}^{-5}$
- D** $2\text{N}_2\text{O}_{(g)} \rightleftharpoons 2\text{N}_{2(g)} + \text{O}_{2(g)}$; $K = 3.5 \times 10^{33} \text{ mol L}^{-1}$

Question

The value of K_c for the reaction

$2A \rightleftharpoons B + C$ is 2×10^{-3} . At a given time, the composition of reaction mixture is $[A] = [B] = [C] = 3 \times 10^{-4}$ M. In which direction the reaction will proceed?

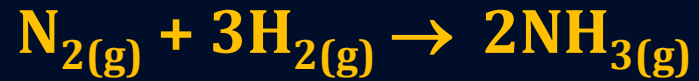
Question

Consider the reaction

$\text{SO}_2\text{Cl}_2(\text{g}) \rightarrow \text{SO}_2(\text{g}) + \text{Cl}_2(\text{g})$; at 375°C , the value of equilibrium constant for the reaction is 0.0032. It was observed that concentration of the three species is 0.050 mol/l each at a certain instant. Discuss what will happen in the reaction vessel?

Question

The equilibrium of the reaction

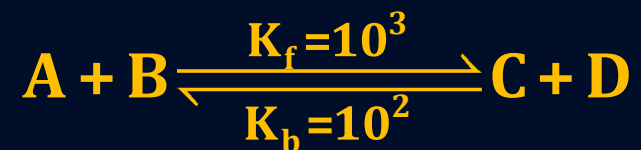


will be shifted to the right when:

- A** $K_p < Q$
- B** $Q < K_p$
- C** $Q = K_p$
- D** $Q < 2K_p$

Question

Consider the following reaction approaching equilibrium at 27°C and 1 atm pressure



The standard Gibb's energy change ($\Delta_r G^\circ$) at 27°C is (-)_____ kJ mol⁻¹ (Nearest integer).

(Given: $R = 8.3 \text{ J K}^{-1} \text{ mol}^{-1}$ and $\ln 10 = 2.3$)

Question

Consider the following reversible chemical reactions:



The relation between K_1 and K_2 is

A $K_1 K_2 = \frac{1}{3}$

B $K_1 = K_1^3$

C $K_2 = K_1^{-3}$

D $K_1 K_2 = 3$

Question

The equilibrium constant K_c for the reaction

$\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$ at 700 K is 49.

What is the equilibrium constant for the reaction

$\text{HI}(\text{g}) \rightleftharpoons \frac{1}{2}\text{H}_2(\text{g}) + \frac{1}{2}\text{I}_2(\text{g})$ at same temperature?

A 49

B 0.005

C 0.143

D 14.3

Question

For the reaction $2\text{NO}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \rightleftharpoons \text{N}_2\text{O}_5(\text{g})$, if the equilibrium constant is K_p , then the equilibrium constant for the reaction would be:



- A** K_p^2
- B** $\frac{2}{K_p}$
- C** $\frac{1}{K_p^2}$
- D** $\frac{1}{\sqrt{K_p}}$

Question

$A + B \rightleftharpoons C + 2D$ equilibrium constant = K_1 , $A + E \rightleftharpoons C + F$ equilibrium constant = K_2

Then equilibrium constant for the reaction $2D + E \rightleftharpoons B + F$ will be:

- A** $\frac{K_2}{K_1}$
- B** $\frac{K_1}{K_2}$
- C** $K_1 + K_2$
- D** $K_1 \times K_2$

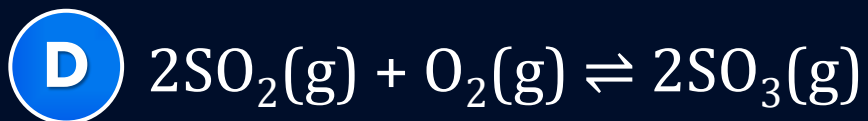
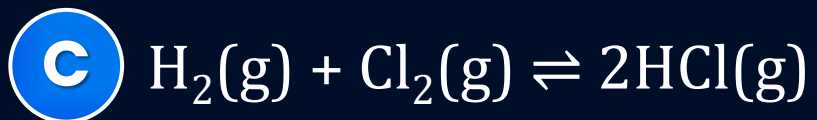
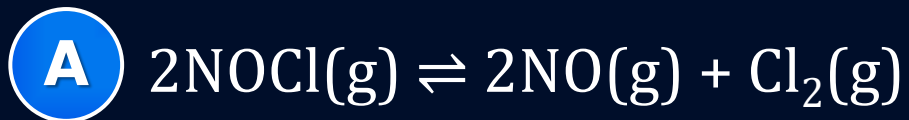
Question

The equilibrium constant (K_p) for the reaction $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$ is 16. If the volume of the container is reduced to one-half of its original volume, the value of K_p for the reaction at the same temperature will be:

- A** 32
- B** 64
- C** 16
- D** 4

Question

For which reaction is $K_p = K_c$:



Question

For a gaseous reaction, $2A + 3B \rightleftharpoons 2C$, which is correct?

A $\log \frac{K_P}{K_C} - \log \frac{1}{RT} = 0$

B $\log \frac{K_P}{K_C} - 3 \log \frac{1}{RT} = 0$

C $\log \frac{K_P}{K_C} + 3 \log \frac{1}{RT} = 0$

D $\log \frac{K_P}{K_C} + 2 \log \frac{1}{RT} = 0$

Question

For $\text{PCl}_{3(g)} + \text{Cl}_{2(g)} \rightleftharpoons \text{PCl}_{5(g)}$, $2K_p = K_c$ is possible
When:

A $T = 6.09 \text{ K}$

B $T = 12.18 \text{ K}$

C $T = 24.36 \text{ K}$

D $T = 25.2 \text{ K}$

Question

Consider the following reaction at 298 K.

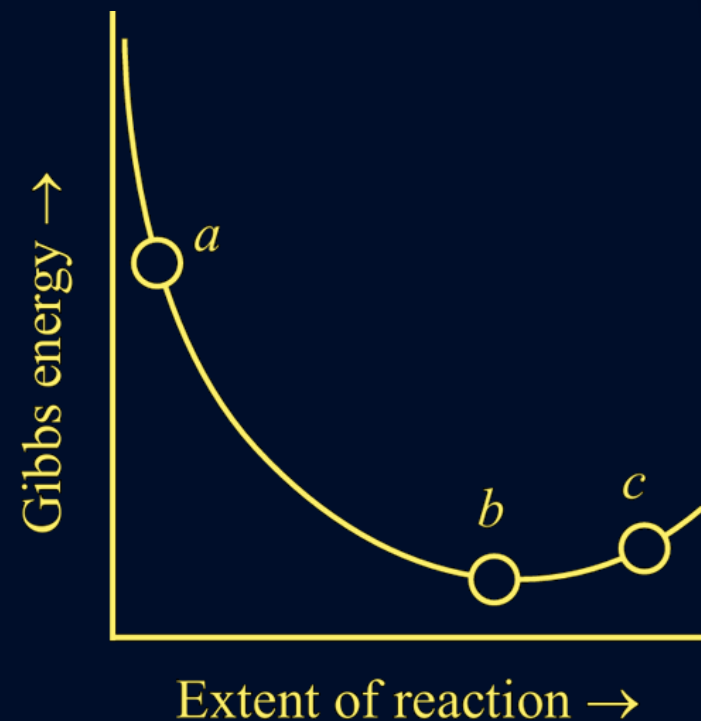


$\Delta_{\text{r}}G^{\ominus}$ for the reaction is kJ.

(Given $R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$)

Question

Consider the graph of Gibbs free energy G vs Extent of reaction. The number of statement/s from the following which are true with respect to points (a), (b) and (c) is...



- A** Reaction is spontaneous at (a) and (b)
- B** Reaction is at equilibrium at point (b) and nonspontaneous at point (c)
- C** Reaction is spontaneous at (a) and nonspontaneous at (c)
- D** Reaction is non-spontaneous at (a) and (b)

Question

The **INCORRECT** match in the following is:

- A** $\Delta G^\circ < 0, K < 1$
- B** $\Delta G^\circ = 0, K = 1$
- C** $\Delta G^\circ > 0, K < 1$
- D** $\Delta G^\circ < 0, K > 1$

Question

3.00 mol of PCl_5 kept in 1L closed reaction vessel was allowed to attain equilibrium at 380K. Calculate composition of the mixture at equilibrium. $K_c = 1.80$

Question

The value of K_p for the reaction, $\text{CO}_2(\text{g}) + \text{C}(\text{s}) \rightarrow 2\text{CO}(\text{g})$ is 3.0 at 1000 K. Initially $P_{\text{CO}_2} = 0.48$ bar and $P_{\text{CO}} = 0$ bar and pure graphite is present, calculate the equilibrium partial pressures of CO and CO_2 .

Question

The value of $K_c = 4.24$ at 800K for the reaction,
 $\text{CO(g)} + \text{H}_2\text{O (g)} \rightarrow \text{CO}_2\text{(g)} + \text{H}_2\text{(g)}$ Calculate equilibrium concentrations of CO_2 , H_2 ,
 CO and H_2O at 800 K, if only CO and H_2O are present initially at concentrations of
0.10M each.

Question

At 600 K, 2 mol of NO are mixed with 1 mol of O₂.



The reaction occurring as above comes to equilibrium under a total pressure of 1 atm. Analysis of the system shows that 0.6 mol of oxygen are present at equilibrium. The equilibrium constant for the reaction is _____. (Nearest integer)

Question

Reaction between N_2 and O_2 takes place as follows:

$2\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{N}_2\text{O}(\text{g})$. If a mixture of 0.482 mol N_2 and 0.933 mol of O_2 is placed in a 10 L reaction vessel and allowed to form N_2O at a temperature for which $K_c = 2.0 \times 10^{-37}$, determine the composition of equilibrium mixture.

Question

Nitric oxide reacts with Br_2 and gives nitrosyl bromide as per reaction given below:



When 0.087 mol of NO and 0.0437 mol of Br_2 are mixed in a closed container at constant temperature, 0.0518 mol of NOBr is obtained at equilibrium. Calculate the equilibrium amount of NO and Br_2 .

Question

A vessel at 1000 K contains CO_2 with a pressure of 0.5 atm. Some of CO_2 is converted into CO on addition of graphite. If total pressure at equilibrium is 0.8 atm, then K_p is:

- A** 3 atm
- B** 1.8 atm
- C** 0.18 atm
- D** 0.3 atm

Question

4.5 moles each of hydrogen and iodine is heated in a sealed ten litre vessel. At equilibrium, 3 moles of HI were found. The equilibrium constant for $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$ is

Question



In an experiment, 2.0 mole of NOCl was placed in a one-litre flask and the concentration of NO after equilibrium established, was found to be 0.4 mol/L. The equilibrium constant at 30°C is _____ $\times 10^{-4}$

Question

A sample of HI(g) is placed in flask at a pressure of 0.2 atm. At equilibrium the partial pressure of HI(g) is 0.04 atm. What is K_p for the given equilibrium?



Question

For the reaction, $A + B \rightleftharpoons C + D$, If initially concentration of A and B both are equal but at equilibrium concentration of D is twice then that of A, then, what will be the equilibrium constant of reaction?

- A** $4/9$
- B** $9/4$
- C** $1/9$
- D** 4

Question

A mixture of 0.3 mole of H_2 and 0.3 mole of I_2 is allowed to react in a 10 lit evacuated flask at 500°C . The reaction is $\text{H}_2 + \text{I}_2 \rightleftharpoons 2\text{HI}$, the K is found to be 6. The amount of unreacted I_2 at equilibrium is

- A** 0.15 mole
- B** 0.06 mole
- C** 0.03 mole
- D** 0.2 mole

Question

28 g of N_2 and 6 g of H_2 were kept at 400°C in 1 litre vessel, the equilibrium mixture contained 27.54 g of NH_3 . The approximate value of K_c for the reaction

$\text{N}_{2(g)} + 3\text{H}_{2(g)} \rightleftharpoons 2\text{NH}_{3(g)}$ can be

- A** 75
- B** 50
- C** 25
- D** 100

Question

In the reaction $A + 2B \rightleftharpoons 2C$, if 2 moles of A, 3 moles of B and 2 moles of C are placed in a 2 L flask and the equilibrium concentration of C is 0.5 mol/L. The equilibrium constant (K_c) for the reaction is

- A** 0.073
- B** 0.147
- C** 0.05
- D** 0.026

Question

A reaction is $A + B \rightleftharpoons C + D$. Initially we start with equal concentration of A and B. At equilibrium the moles of C is two times of A. What is the equilibrium constant of the reaction?

- A** 4
- B** 2
- C** $1/4$
- D** $1/2$

Question

PCl_5 dissociates as $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$

5 moles of PCl_5 are placed in a 200 litre vessel which contains 2 moles of N_2 and is maintained at 600 K. The equilibrium pressure is 2.46 atm. The equilibrium constant K_p for the dissociation of PCl_5 is _____ $\times 10^{-3}$ g. (nearest integer)

(Given $R = 0.082 \text{ L atm K}^{-1} \text{ mol}^{-1}$; Assume ideal gas behaviour)

Question

4.0 moles of argon and 5.0 moles of PCl_5 are introduced into an evacuated flask of 100 litre capacity at 610 K. The system is allowed to equilibrate. At equilibrium, the total pressure of mixture was found to be 6.0 atm. The K_p for the reaction is. [Given $R = 0.082 \text{ L atm K}^{-1} \text{ mol}^{-1}$]

A 2.25

B 6.24

C 12.13

D 15.24

Question

In the beginning of the reaction, $A \rightleftharpoons B$, 4 moles of A are taken, out of which 0.8 moles gets dissociated. What is the degree of dissociation of A?

- A** 1
- B** 0.125
- C** 0.2
- D** 3

Question

4 mol of PCl_5 are heated at constant temperature in closed container. If degree of dissociation for PCl_5 is 0.5 then calculate total number of moles at equilibrium

- A** 4.5
- B** 6
- C** 3
- D** 4

Question

At a certain temp. $2\text{HI} \rightleftharpoons \text{H}_2 + \text{I}_2$ Only 50% HI is dissociated at equilibrium. The equilibrium constant is

- A** 0.25
- B** 1.0
- C** 3.0
- D** 0.50

Question

For the reaction, $\text{N}_2\text{O}_{3(g)} \rightleftharpoons \text{NO}_{(g)} + \text{NO}_{2(g)}$; total pressure = P, Degree of dissociation = 50%. Then K_p would be

- ☐ A $3P$
- ☐ B $2P$
- ☐ C $\frac{P}{3}$
- ☐ D $\frac{P}{2}$

Question

Consider the following chemical equilibrium of the gas phase reaction at a constant temperature : $A(g) \rightleftharpoons B(g) + C(g)$

If p being the total pressure, K_p is the pressure equilibrium constant and α is the degree of dissociation, then which of the following is true at equilibrium?

- A** If p value is extremely high compared to K_p , $\alpha \approx 1$
- B** When p increases α decreases
- C** If K_p value is extremely high compared to p , α becomes much less than unity
- D** When p increases α increases

Question

The equilibrium constant (K_p) for the decomposition of gaseous H_2O



is related to degree of dissociation (α) at a total pressure p , given by:

A $K_p = \frac{\alpha^3 p^{1/2}}{(1 + \alpha)(2 + \alpha)^{1/2}}$

B $K_p = \frac{\alpha^3 p^{3/2}}{(1 - \alpha)(2 + \alpha)^{1/2}}$

C $K_p = \frac{\alpha^{3/2} p^2}{(1 - \alpha)(2 + \alpha)^{1/2}}$

D $K_p = \frac{\alpha^{3/2} p^{1/2}}{(1 - \alpha)(2 + \alpha)^{1/2}}$

Question

At the equilibrium of the reaction,



the observed molecular weight of N_2O_4 is 80 g mol^{-1} at 350 K . The percentage of dissociation of $\text{N}_2\text{O}_4 (\text{g})$ at 350 K is:

- A** 10%
- B** 15%
- C** 20%
- D** 18%

Question

At 250°C and 1 atmospheric pressure, the vapour density of PCl_5 is 57.9. What will be the dissociation of PCl_5 ?

- A** 1.00
- B** 0.90
- C** 0.80
- D** 0.65

Question

2 moles of PCl_5 are heated in a closed vessel of 2 litre capacity. At equilibrium 40% of PCl_5 is dissociated into PCl_3 and Cl_2 . The value of equilibrium constant is

- A** 0.266
- B** 0.53
- C** 2.66
- D** 5.3

Question

2 moles of PCl_5 are heated in a closed vessel of 2 litre capacity. At equilibrium 40% of PCl_5 is dissociated into PCl_3 and Cl_2 . The value of equilibrium constant is

- A** 0.266
- B** 0.53
- C** 2.66
- D** 5.3

Question

2 mol of N_2 is mixed with 6 mol of H_2 in a closed vessel of one litre capacity. 50% of N_2 is converted into NH_3 at equilibrium, the value of K_c for the reaction $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$ is:

- A** $4/27$
- B** $27/4$
- C** $1/27$
- D** 24

Question

Two moles of HI were heated in a sealed tube at 440°C till the equilibrium was reached. HI was found to be 22% decomposed. The equilibrium constant for dissociation is

- A** 0.282
- B** 0.0796
- C** 0.0199
- D** 1.99

Question



If the degree of dissociation and initial concentration of both the reactants $X(g)$ and $A(g)$ are equal, then the ratio of the total pressure at equilibrium $\left(\frac{p_1}{p_2}\right)$ is equal to $x : 1$. The value of x is _____ (Nearest integer)

Question

N_2O_4 dissociates as $\text{N}_2\text{O}_{4(\text{g})} \rightleftharpoons 2\text{NO}_{2(\text{g})}$ at 273 K and 2 atm pressure. The equilibrium mixture has a density of 41. What will be the degree of dissociation

- A** 14.2%
- B** 16.2%
- C** 12.2%
- D** None

Question

Le-Chatlier principle is not applicable to

- A** $\text{H}_{2(g)} + \text{I}_{2(g)} \rightleftharpoons 2\text{HI}_{(g)}$
- B** $\text{Fe}_{(s)} + \text{S}_{(s)} \rightleftharpoons \text{FeS}_{(s)}$
- C** $\text{N}_{2(g)} + 3\text{H}_{2(g)} \rightleftharpoons 2\text{NH}_{3(g)}$
- D** $\text{N}_{2(g)} + \text{O}_{2(g)} \rightleftharpoons 2\text{NO}_{(g)}$

Question

For the reaction: $A + B + \text{Heat} \rightarrow C + D$, if the temperature is increased, the concentration of the products will

- A** Increase
- B** Decrease
- C** Remain same
- D** Become zero

Question

Which of the following equilibrium is not shifted by increase in the pressure?

- A** $\text{H}_{2(g)} + \text{I}_{2(g)} \rightleftharpoons 2\text{HI}_{(g)}$
- B** $\text{N}_{2(g)} + 3\text{H}_{2(g)} \rightleftharpoons 2\text{NH}_{3(g)}$
- C** $2\text{CO}_{(g)} + \text{O}_{2(g)} \rightleftharpoons 2\text{CO}_{2(g)}$
- D** $2\text{C}_{(g)} + \text{O}_{2(g)} \rightleftharpoons 2\text{CO}_{(g)}$

Question

Consider the following equation: $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$, $\Delta H = -190 \text{ kJ}$

The number of factors which will increase the yield of SO_3 at equilibrium from the following is _____

- A. Increasing temperature
- B. Increasing pressure
- C. Adding more SO_2
- D. Adding more O_2
- E. Addition of catalyst

Question

In the following system,

$\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$ at equilibrium, upon addition of xenon gas at constant T and P, the concentration of

- A** PCl_5 will increase
- B** Cl_2 will decrease
- C** PCl_5 , PCl_3 and Cl_2 remain constant
- D** PCl_3 will increase

Question

Given below are two statements

Statement I: A catalyst cannot alter the equilibrium constant (K_c) of the reaction at constant temperature remaining constant.

Statement II: A homogenous catalyst can change the equilibrium composition of the reaction system at constant temperature remaining constant.

In the light of the above statements, choose the correct answer from the options given below

- A** Statement I is false but Statement II is true
- B** Both Statement I and Statement II are true
- C** Both Statement I and Statement II is false
- D** Statement I is true but Statement II is false

Question

Consider the equilibrium $\text{CO(g)} + 3\text{H}_2\text{(g)} \rightleftharpoons \text{CH}_4\text{(g)} + \text{H}_2\text{O(g)}$

If the pressure applied over the system increases by two fold at constant temperature then

- (A) Concentration of reactants and products increases.
- (B) Equilibrium will shift in forward direction.
- (C) Equilibrium constant increases since concentration of products increase.
- (D) Equilibrium constant remains unchanged as concentration of reactants and products remain same.

Choose the correct answer from the options given below :

- | | |
|--------------------------------|--------------------------------|
| A (A) and (B) only | B (B) and (C) only |
| C (A), (B) and (C) only | D (A), (B) and (D) only |

Question

The following reaction occurs in the Blast furnace where iron ore is reduced to iron metal



Using the Le-Chatelier's principle, predict which one of the following will not disturb the equilibrium

- A** Addition of Fe_2O_3
- B** Addition of CO_2
- C** Removal of CO
- D** Removal of CO_2

Question

Formation of SO_3 takes place according to the reaction



Which of the following factors favours the formation of SO_3 ?

- A** Increase in temperature
- B** Increase in pressure
- C** Removal of oxygen
- D** Increase in volume

Question

Which one of the following condition will favour maximum formation of the product in the reaction,



- A** Low temperature and high pressure
- B** Low temperature and low pressure
- C** High temperature and high pressure
- D** High temperature and low pressure

Question

Which of the following reactions will get affected by increasing the pressure? Also, mention whether change will cause the reaction to go into forward or backward direction.



Question

At 473 K, equilibrium constant K_c for decomposition of phosphorus pentachloride, PCl_5 is 8.3×10^{-3} . If decomposition is depicted as,



Write an expression for K_c for the reaction.

What is the value of K_c for the reverse reaction at the same temperature?

What would be the effect on K_c if a) More PCl_5 is added b) pressure is increased c) The temperature is increased?

Question

Dihydrogen gas is obtained from natural gas by partial oxidation with steam as per following endothermic reaction:



Write an expression for K_p for the above reaction.

How will the values of K_p and composition of equilibrium mixture be affected by (i) increasing the pressure (ii) increasing the temperature (iii) using a catalyst?